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1



required to control transmission effectively, a gap that is highly magnified when deployed against a syndemic backdrop.

Additionally, the necessity of an optimized, high-affinity chimeric construct becomes predominant. To transition such a multivalent design from *in silico* parameters to a functional mRNA-LNP delivery platform, precise computational modeling must guide the molecular cloning process. The implementation of SnapGene serves as a vital bridge in this phase, enabling the proactive simulation of the chimeric MpoxHIV sequence's insertion into a plasmid template for *in vitro* transcription (IVT). Replacing the MVA vector with an mRNA-Lipid Nanoparticle (LNP) platform mitigates the risk of vector-induced immunity, preventing instances where the viral vector itself triggers an immune response that invalidates the target vaccine antigens. By utilizing SnapGene to visually map specific restriction sites, specifically BamHI and HindIII, and simulate *in silico* polymerase chain reaction (PCR) amplification, the structural preservation of the essential AAY, GPGPG, and KK linkers can be guaranteed at critical genomic boundaries before entering the wet lab. This rigorous cloning simulation directly builds upon the foundational framework proposed by Ahmad et al. (2025), who developed the foundational dual-pathogen immunoinformatic targeting of Mpox and HIV proteins. To overcome the systemic immunogenic bottlenecks of viral vectors identified by Zaeck et al. (2023), this study shifts to an mRNA-LNP delivery mechanism to maximize downstream neutralizing efficacy.

Immunoinformatics-based multi-epitope vaccine design has been widely used to combat viral pathogens individually, for instance, targeting Mpox surface-binding proteins (Yousaf et al., 2022) and the HIV-1 gp120 envelope (Naveed et al., 2023). Structural stability and immunogenicity in these constructs are known to depend on strategic linker selection (AAY for Cytotoxic T-Lymphocyte (CTL) epitope processing, GPGPG to prevent junctional neo-epitopes in Helper T-Lymphocyte (HTL) sequences, the KK linker is uniquely suited for connecting B-cell epitopes to maintain appropriate independent conformation and folding) and adjuvant use, particularly β -defensin 3, often connected via the rigid EAAAK linker which enhances innate/adaptive responses via TLR2/TLR4 signaling (Rcheulishvili et al., 2023; Naveed et al., 2022; Yousaf et al., 2022; Tani et al., 2000; Ahmed et al., 2025). However, the literature presents varying findings regarding the efficacy of these designs in co-infected populations. Advances in reverse vaccinology and immunoinformatics provide a promising alternative. Computational tools enable the systematic identification of high-affinity, non-toxic, and non-allergenic epitopes, thereby refining candidate selection prior to experimental validation. Yousaf et al. (2022) designed a peptide-based Mpox vaccine candidate targeting viral surface proteins with strong predicted immunogenicity. Similarly, Naveed et al. (2022; 2023) developed multi-epitope constructs targeting conserved regions of HIV-1 gp120, gp41, and Gag proteins. These studies demonstrate the potential of *in silico* screening to streamline vaccine development. In multi-epitope vaccines, structural configuration is critical; linkers such as AAY and GPGPG facilitate antigen processing and epitope presentation, as shown by Rcheulishvili et al. (2023). Molecular

adjuvants such as β -defensin 3 further enhance immune signaling, particularly through Toll-like receptor pathways including TLR-2/4 interactions, strengthening the projected immunogenic profile of chimeric constructs.

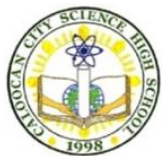
While *in silico* models predict high global population coverage and strong binding affinities, clinical data from traditional orthopoxvirus vaccines show reduced immunogenicity in patients with advanced HIV (CD4 counts below 200 cells/mm³) (Ogaya et al., 2026; Taha et al., 2024). Clinically, the syndemic interaction between HIV and Mpox is also documented: advanced HIV infection correlates with more severe Mpox outcomes (Mitjà et al., 2023). Locally, Mpox cases in the Philippines are predominantly Clade II, and HIV-1 CRF01_AE accounts for roughly 77% of cases (Salvaña et al., 2022).

Consequently, while individual viral components and linker strategies have been studied, limited research has examined the immunogenicity of a chimeric vaccine specifically tailored to Philippine-prevalent strains such as CRF01_AE. The urgency of addressing this syndemic is influenced by the epidemiological context of the Philippines, where identified Mpox cases are classified as Clade II (Ogaya et al., 2026), and the HIV-1 CRF01_AE strain represents approximately 77% prevalence (Salvaña et al., 2022).

Despite this body of work, no study has combined HIV-1 and Mpox epitopes into a single chimeric construct validated against Philippine-prevalent strains. Existing dual-pathogen vaccine models rarely integrate this local strain dominance into epitope selection or receptor-binding simulations. This underscores a significant gap in current research. Existing multi-epitope designs treat the two pathogens independently (Mitjà et al., 2023), and universal orthopoxvirus vaccine models, while computationally promising, have not been docked or optimized against strains circulating in the Philippine population (Rcheulishvili et al., 2023). Consequently, there is no localized, dual-purpose immunization model tailored to populations at elevated risk of complications (CROI, 2023).

The rising HIV-1 CRF01_AE and Mpox Clade II co-infections in the Philippines overwhelm healthcare, amplifying severity in immunocompromised individuals, yet no targeted, region-specific vaccine exists. General orthopoxvirus shots ignore local strains and molecular needs for robust immunity. While Biswas et al. (2026) and Tan et al. (2023) have explored antigenic targets and the biological/social interaction between the two diseases, an integrated preventive strategy — one that screens epitopes from both pathogens under a single antigenicity, toxicity, and immunogenicity pipeline, and validates the resulting construct through docking against locally dominant strains — remains unexplored.

To combat this syndemic biological problem, the study's goal is to design and evaluate the binding affinity and predicted immunogenicity of a chimeric multi-epitope vaccine targeting the HIV-Mpox syndemic. Using an *in silico* framework that screens candidate epitopes from Mpox (L1R, B5R, A35R) and HIV-1 (gp120, gp41, p24, p17) proteins for antigenicity,



conservation, toxicity, allergenicity, and population coverage. The construct integrates AAY, GPGPG, and KK linkers with a β -defensin 3 adjuvant to maintain physicochemical stability, and its performance is evaluated through ExPASy ProtParam (RMSD/RMSF) and Monte Carlo-based immune simulation (C-ImmSim) over a three-dose regimen. Furthermore, to verify This study aims to design and evaluate a multi-epitope vaccine construct that targets conserved proteins of the Mpox virus and HIV-1, as well as to model its interaction within the immune system. The goal is to establish an in silico framework for dual-pathogen immunization. As follows, the research aims to answer the following questions:

1. Which T-cell (MHC-I and MHC-II) and B-cell epitopes derived from the selected HIV-1 (gp120, gp41, p24, and p17) and Mpox (L1R, B5R, and A35R) proteins satisfy the criteria for antigenicity, conservancy, population coverage, non-toxicity, and non-allergenicity?
2. What architectural arrangement of these epitopes using specific linkers (AAY, GPGPG, and KK) and the β -defensin 3 adjuvant ensures a stable, chimeric vaccine construct with high physicochemical stability and favorable protein folding is most optimal?
3. To what extent does the predicted tertiary structure meet standard stereochemical quality thresholds across global metrics (MolProbity score, Ramachandran-favored distribution) and local geometry parameters (rotamer rotamericity, C α geometry, and CaBLAM outliers), and how are local stereochemical deviations spatially distributed between the rigid adjuvant domain and flexible epitope linkers??
4. What are the molecular docking characteristics of the vaccine construct's high-confidence adjuvant domain with human Toll-Like Receptor 2 (TLR-2) and Toll-Like Receptor 4 (TLR-4) in terms of binding affinity and docking scores?
5. How does the adjuvant-receptor complex behave dynamically over a 100-ns Molecular Dynamics (MD) simulation in terms of structural deviation (RMSD), flexibility (RMSF), and compactness (Radius of Gyration)?
6. What are the predicted structural flexibility and deformability characteristics of the adjuvant-receptor complex in terms of eigenvalues, covariance maps, deformability profiles, and B-factor analyses?
7. What are the predicted codon optimization and expression characteristics of the vaccine construct based on Codon Adaptation Index (CAI), GC content, and in silico IVT plasmid template cloning?
8. Based on comparative C-ImmSim probabilistic simulations, do the independent HIV and Mpox components produce proportional immune

responses to verify that neither construct overshadows the other?

9. What is the predicted immunogenic performance of the chimeric designed vaccine following a simulated three-dose mRNA-LNP regimen in C-ImmSim, modeled under delayed-onset antigen availability parameters, in terms of antibody production (IgM and IgG), cytokine response (IFN- γ), and T-cell clonal diversity as measured by the Simpson diversity index?

The primary purpose of this research is to design and computationally validate a chimeric multi-epitope vaccine capable of addressing the syndemic relationship between the Human Immunodeficiency Virus (HIV-1) and the Mpox virus. As established by Mitjà et al. (2023), the biological interaction between these two pathogens significantly increases mortality rates, and implementing a vaccine strategy that is safe for immunocompromised populations. Furthermore, the paper's construct of an optimized vaccine sequence using beta-defensin 3 as an adjuvant and specialized linkers to maintain structural stability and physicochemical viability (Rcheulishvili et al., 2023), and an mRNA-LNP delivery platform bypasses vector-induced immunogenic interference. With a critical component of this study being the evaluation of the vaccine's binding affinity toward Toll-like Receptor 2 (TLR-2) and Toll-like Receptor 4 (TLR-4) through molecular docking to ensure the initiation of a robust innate immune response (Yousaf et al., 2022). Through C-ImmSim immune simulations, this study aims to model the efficacy of a three-dose immunization regimen and comparatively validate the independent components against immunodominance, ultimately providing a theoretical blueprint for a dual-purpose vaccine that mitigates the lethal cooperation between these two pathogens.

Beyond structural design, the research aims to contribute to the existing literature by outlining a framework for dual-pathogen epitope selection. The resulting data and simulations may serve as a reference point for future experimental validation and clinical investigation. By tailoring epitope selection to CRF01_AE and Clade II strains prevalent in high-density urban settings such as Metro Manila, this research offers a computationally validated blueprint intended to guide future experimental and clinical investigation into dual-pathogen immunization strategies.

This research is strictly delimited to the in silico computational design and characterization of a multi-epitope chimeric vaccine targeting HIV-1 and Mpox. Additionally, it only considered the use of the β -defensin 3 as an adjuvant, an mRNA-LNP delivery platform, and the AAY, GPGPG, and KK as linkers, excluding other possible configurations. The scope of the study focuses on seven specific viral proteins: L1R, B5R, and A35R for Mpox, and gp120, gp41, p24, and p17 for HIV-1. Conversely, this study does not involve in vitro (laboratory) or in vivo (animal/human) trials. The results are intended solely as a computationally validated blueprint. A significant limitation of this study is its reliance on the predictive accuracy of bioinformatic algorithms and the static nature of protein

databases, which represent high-probability models rather than guaranteed biological outcomes. Furthermore, the 100ns simulation window provides a microscopic snapshot of molecular behavior and may not capture long-term structural changes that occur beyond this timeframe.

METHODOLOGY

Research Design

This study utilizes a quantitative, in silico experimental research design. It involves the computational screening of viral proteomes (proteins and genomes) to design a chimeric vaccine, followed by the simulation of molecular interactions to evaluate binding affinity, structural stability, and probabilistic immune modelling to predict random immune cell interactions with the vaccine.

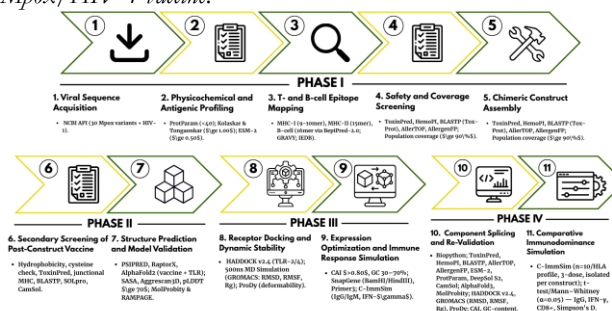
Research Instrument

The study utilized Python version 3.14 to perform simulations and fulfill the computational requirements necessary for the paper. The research used Biopython as the foundation for handling and reading DNA sequences. For the vaccine construction, the NCBI API was used to retrieve FASTA-formatted sequences directly into the script. Solubility and Physicochemical Stability tests included the utilization of libraries and tools such as Bio.SeqUtils.ProtParam, CamSol, DeepSol S2, and Grand Average of Hydropathicity (GRAVY), whilst 3-dimensional structural solubility analyses were conducted by Aggrescan3D, CamSol structural mode, PDB2PQR, APBS, DeepSol S2, and FoldX 5, FreeSASA 2.2.1. Antigenicity tests were performed with the employment of the Kolaskar & Tongaonkar Antigenicity Scale, and Evolutionary Scale Modeling 2 (ESM-2) (Agarwal et al., 2021; Furui et al., 2025), while epitope selection for the identification of candidate k-mer sequences for MHC-I, MHC-II, and B-cells was performed by utilizing a Sliding-window k-mer generator (Shabbir et al., 2025). The filtration process to identify which MHC I and II candidates were appropriate was screened through the IEDB MHC Tools API (https://tools-cluster-interface.iedb.org/tools_api/mhci/), and BigMHC. For the selection of the B-cell peptides, BepiPred-2.0 and SEMA 2.0 were incorporated. After selection, it was then followed by a GRAVY filter using the Kyte–Doolittle hydropathy scale. Toxicity tests were enforced with the use of their Cysteine count, the Hydrophobic Ratio Filter, ToxinPred, HemoPI, Short-peptide homology check: Basic Local Alignment Search Tool for Proteins (BLASTP), BLAST+ 2.16.0, and the UniProt Tox-Prot subset. Allergenicity screening included the application of Q/N Fraction, Charged Residue Count, AllerTop, and AllergenFP. The prediction for the Human Leukocyte Antigen (HLA) alleles for MHC I and II was conducted by MHCflurry and MHCnuggets, respectively. The Population Coverage Analysis test also utilized allele-frequency tables downloaded from AFND (allelefrequencies.net) Vaccine construction was done using

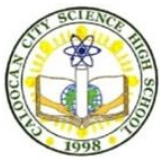
Biopython (SeqRecord). Additionally, the IEDB Query API (query-api.iedb.org) curated the epitope and assay records

For the tertiary and secondary structure of the protein sequence, 3D Model creation tools like AlphaFold3 (<https://alphafoldserver.com/welcome>), and PsiPred (<https://bioinf.cs.ucl.ac.uk/psipred>) and NetSurfP-3.0 (services.healthtech.dtu.dk/services/NetSurfP-3.0/), pydssp 0.9.1 were used, respectively. MolProbity (<http://molprobity.biochem.duke.edu/>) was employed to perform the Ramachandran Plot and present the plot. Toll-like receptors (TLR2 and TLR4) were retrieved from the RCSB REST API (data.rcsb.org). Phenix, PDBFixer, and molprobity.cablam were incorporated to repair the model and for backbone validation. Energy minimization was conducted by OpenMM 8.6. Fold-confidence analysis was incorporated by Bio.PDB.Superimposer. Molecular Docking was conducted using Haddock v2.4 (<https://wenmr.science.uu.nl/haddock2.4/>), and a 100ns MD Simulation was performed through GROMACS (<https://www.gromacs.org/>) with the utilization of the DOST-COARE High-Performance Computing (HPC). Additionally, Molecular Dynamic analysis of the results of the simulation was managed by the MDAnalysis library. Structural Flexibility analysis of the vaccine was directed by ProDy, while Codon Optimization was conducted with the tools of Codon Adaptation Index (CAI) (<https://ppuigbo.me/programs/CAIcal/>), GC Content Calculator (<https://en.vectorbuilder.com/tool/gc-content-calculator.html>), and with Snap Gene (<https://www.snapgene.com/>) constructed the IVT template, RNAfold (ViennaRNA Package) for the prediction of secondary mRNA folding structure and calculate the minimum free energy (G), DeepTMHMM and SignalP 6.0 (Updated Expression Profiling Suite) for sequence refinement, and In Silico Microfluidic Parameter Calculators (SHM Modeling) to simulate advection parameters, establishing specific flow rate ratios, and charge ratios. Additionally, an in silico Polymerase Chain Reaction (PCR) was conducted with Primer3 (<https://primer3.ut.ee/>). Lastly, Probabilistic Immune Simulations were done with C-ImmSim (<https://kraken.iac.rm.cnr.it/C-IMMSIM/index.php>).

Figure 1
Four phase computational workflow for the design of chimeric Mpox/HIV-1 vaccine.



Note. In this figure, the sequential 11-step workflow is divided into four phases. The bulleted items list the specific



bioinformatics tools, servers, and evaluation thresholds applied at each stage.

Data Gathering Procedure

I. Epitope Selection and Construction

A. Sequence Retrieval

To capture global sequence diversity across target antigens, up to 30 non-redundant protein variants were targeted for each antigen from NCBI GenBank. Full target counts (30 sequences each) were retrieved for HIV-1 gp120, gp41, p17, and p24, as well as Mpox L1R. For Mpox A35R and B5R, complete non-redundant database retrieval yielded 27 and 25 variants, respectively, resulting in a total dataset of 202 sequences across all seven target antigens. Due to inconsistent clade metadata annotations in GenBank records for A35R and B5R, sequence filtering accommodated both explicitly annotated and inferred Clade II isolates to maximize sequence inclusion while maintaining lineage relevance. (Shabbir et al., 2025).

B. Stability Testing

For physicochemical analysis of the selected proteins, the study considered that attaining a score below 40 from the Bio.SeqUtils.ProtParam library would be stable for a laboratory setting (Savojardo et al., 2024).

C. Antigenicity Screening

To filter out the most viable epitopes of both Clade IIa and Clade IIb, as well as for HIV-1 CRF01_AE. The study used a propensity mean threshold of 1.00 and a window of 7 residues on windowed K&T scores to indicate antigenic epitopes. The Kolaskar & Tongaonkar scale used is represented below (Odhar, 2020).

Table 1.

Kolaskar & Tongaonkar per-residue antigenic propensities

Kolaskar & Tongaonkar scale				
'A': 1.8	'C': 1.412	'D': 0.866	'E': 0.851	'F': 1.091
'G': 0.874	'H': 1.105	'I': 1.152	'K': 0.930	'L': 3.8,
'M': 1.126	'N': 0.851	'P': 1.064	'Q': 1.010	'R': 0.873
'S': 1.012	'T': 0.909	'V': 1.187	'W': 1.085	'Y': 1.255

Sequences containing non-standard letters (X, B, Z, J, U, gaps, and stops) were removed before scoring. Only the 20 letters that appeared in the scale were utilized to measure its antigenicity. Per-residue Kolaskar & Tongaonkar propensities (Table 1) were averaged using a sliding window of 7 residues,

wherein the average was assigned to the central residue. Contiguous flagged positions were merged into candidate peptides with a minimum length = 6 aa (Odhar, 2020).

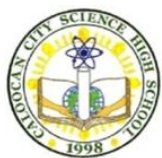
Additionally, to identify highly antigenic proteins suitable for vaccine development, the study utilized the Evolutionary Scale Modeling 2 (ESM-2) protein language model. The classifier generated a probability score ranging from 0.00 to 1.00, representing the predicted likelihood that a protein possesses antigenic characteristics. The threshold of ≥ 0.50 was validated against a curated benchmark dataset (390 antigenic proteins from IEDB; 384 non-antigenic housekeeping proteins) using ROC curve analysis, achieving sensitivity $\geq 85\%$, specificity $\geq 80\%$, PPV ≥ 0.85 , NPV ≥ 0.80 , and AUC > 0.85 , confirming robust discriminatory performance. Proteins with ESM-2 scores ≥ 0.50 were classified as probable antigens and retained for epitope prediction; those < 0.50 were excluded. (Lin et al., 2023; Furui et al., 2025). Consensus agreement between ESM-2 and the Kolaskar & Tongaonkar scale determined final classification, with disagreements flagged for exclusion.

While these dual evaluation procedures were executed as outlined, Section I.C functioned as a cross-methodological consistency check to confirm baseline antigenicity across all source sequences rather than a selective elimination filter.

D. Identification of T and B-cell Epitopes

Candidate epitopes were selected based on their k-mer peptide length. The T-cell epitope selection was based on their Major Histocompatibility Complex (MHC), specifically for MHC Class I (Cytotoxic T-cells), which focuses on 9- to 10-mer peptides, and MHC Class II (Helper T-cells), which focuses on 15-mer peptides. B cells were also considered with k-mer peptide lengths of 16-mer peptides (Shabbir et al., 2025).

The threshold for MHC Class I consisted of the top 1% of binders (1.00%), while for MHC Class II, it included the top 10% of binders (10.00%) (Shabbir et al., 2025). B-cell linear epitope prediction and prioritization were performed with BepiPred-2.0 (Galanis et al., 2021). Each 16-mer was assigned (i) mean_BepiPred (the arithmetic mean of the 16 per-residue scores) and (ii) pct_above (the percentage of residues in the 16-mer) with a BepiPred score ≥ 0.50 (BepiPred-2.0 default) (Galanis et al., 2021). High-tiered prioritized peptides were defined with scores of mean_BepiPred ≥ 0.60 and pct_above $\geq 75\%$, medium-tiered prioritized peptides were defined with scores of mean_BepiPred ≥ 0.50 AND pct_above $\geq 50\%$ and deprioritized



peptides were described with a score of mean_BepiPred ≥ 0.45 AND pct_above $\geq 37.5\%$ (Shabbir et al., 2025).

Additionally, because strong predicted MHC-I binding affinity does not by itself guarantee that a peptide-MHC complex will be recognized by a cognate T-cell receptor, MHC Class I candidates surviving the top-1% binding-percentile threshold (Section I.D) were additionally screened using BigMHC, a deep neural network ensemble trained on mass spectrometry-derived MHC ligand data and transfer-learned on antigen-specific immunogenicity assay data, to estimate T-cell recognition likelihood independent of binding affinity alone (Albert et al., 2023). Peptides receiving a high BigMHC immunogenicity score were retained as prioritized candidates; peptides with a negative score were deprioritized but not automatically excluded, given that low-immunogenicity predictions carry higher uncertainty than binding-affinity predictions and a purely binding-based candidate pool would otherwise remain available as fallback (Albert et al., 2023). This step was applied only to MHC Class I candidates, as the IEDB Immunogenicity tool is validated specifically for CD8⁺ T-cell epitope contexts and does not have an equivalent validated counterpart for MHC Class II in the current IEDB toolset.

To avoid complications in peptide synthesis, handling, and immunoassays. An average hydropathy filter (GRAVY) was used to deprioritize hydrophobic peptides from the hydrophilic peptides: peptides with GRAVY ≥ 0.2 were deprioritized for synthesis due to increased hydrophobicity and potential aggregation, following common multi-epitope design practice (Mahapatra et al., 2020; Dar et al., 2022) and hydropathy benchmarking studies (Waibl et al., 2022). The table below showcases the metrics utilized in the GRAVY (Waibl et al., 2022).

Table 2.
Kyte–Doolittle hydropathy values (per residue; used to compute GRAVY)

<i>Per-residue Kyte–Doolittle hydropathy score</i>				
'A': 1.8	'C': 2.5	'D': -3.5	'E': -3.5	'F': 2.8
'G': -0.4	'H': -3.2	'I': 4.5	'K': -3.9	'L': 3.8
'M': 1.9	'N': -3.5	'P': -1.6	'Q': -3.5	'R': -4.5
'S': -0.8	'T': -0.7	'V': 4.2	'W': -0.9	'Y': -1.3

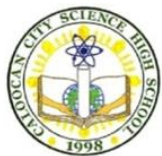
To address the limitation that BepiPred-2.0 predicts only linear B-cell epitopes, while the majority of biologically relevant B-cell epitopes (gp120) are conformational (discontinuous) in nature, native tertiary structures of each source antigen were obtained or predicted prior to chimeric assembly. Solved structures were retrieved from RCSB PDB where available; for proteins lacking a solved structure, AlphaFold3 was used to generate a homology model of the full-length native protein, distinct from the vaccine-construct model generated later in Section II.B. Discontinuous B-cell epitopes were predicted on each native fold using SEMA 2.0, a protein language model-based platform that predicts conformational epitope propensity per residue directly from AlphaFold-predicted or experimentally solved structures, with residues scoring above the platform-recommended epitope probability threshold retained as high-confidence conformational patches (Ivanisenko et al., 2024).

Linear B-cell epitope candidates retained from (BepiPred-2.0, mean_BepiPred ≥ 0.50) were cross-referenced against the SEMA 2.0-derived conformational patches for their respective source protein. Candidates overlapping a conformational patch by $\geq 50\%$ of their residues were classified as structurally corroborated and given priority for inclusion in the final construct, on the basis that these linear stretches are more likely to represent residues that are genuinely surface-accessible and immunologically relevant in the native antigen rather than artifacts of sequence-only prediction (Ivanisenko et al., 2024). Candidates with no conformational-patch overlap were retained as lower-priority linear-only candidates and flagged accordingly in the epitope selection table.

Additionally, the utilization of conservancy benchmarks referenced from the IEDB Conservancy Tool would also be employed to ensure that the selected T and B-cell epitopes are preserved across different viral strains of HIV and Mpox. Benchmarks of $> 20\%$ are minimal conservancy; $> 50\%$ is considered highly acceptable, while 100% is the ideal goal (Waqas et al., 2022; Sanami et al., 2023).

E. Toxicity and Allergenicity Screening

To filter the predicted epitopes to ensure safety for allergenicity and toxicity (Shabbir et al., 2025). Toxicity tests included a Hydrophobic Ratio Filter and Cysteine count, where a sequence having $> 80\%$ hydrophobic residues (A, V, I, L, M, F, W, and Y) which was defined by the formula of $\text{Hydrophobic_Fraction} = (\text{count of A, V, I, L, M, F, W, Y}) / \text{peptide_length}$ or a sequence having more than 2 Cysteines was considered unqualified for the toxicity assessment and was flagged as



“TOXIC” (Sanami et al., 2023). In addition, the peptides were analyzed with sequence tests such as ToxinPred, Hemolysis predictor, and Short-peptide homology check: BLASTP (Sharma et al., 2022). Sequences that were identified as “toxin” by ToxinPred, hemolysis probabilities that were ≥ 0.50 and concurrent with a Hydrophobic_Fraction > 0.60 , and BLASTP sequences that were run against the UniProt Tox-Prot subset that had met all three criteria: percent identity $\geq 80\%$, alignment coverage $\geq 80\%$ (alignment_length / peptide_length), and E-value $\leq 1e-5$ had all been flagged as “TOXIC” (Shabbir et al., 2025). Epitopes that were flagged with “TOXIC” were all removed from the list of possible candidates (Sanami et al., 2023).

While for the allergenicity tests, the Q/N Fraction, which equates to $QN_Ratio = (\text{count}(Q) + \text{count}(N)) / \text{peptide_length}$ (k-mer) where the numerator of the ratio corresponds to the count of glutamine (Q) and asparagine (N) residues, and the denominator represents the peptide length of the sequence (Shabbir et al., 2025). The threshold of a QN_Ratio value > 0.30 flags the sequence as allergenic (Waqas et al., 2023). Additionally, the Charged Residue Count, which deprioritizes sequences that have amino acids with ionizable side chains (D, E, H, and K), included a cutoff of a sequence having more than 4 of the undesired letters (Surface_Charge value > 4) (Shabbir et al., 2025). For bioinformatic prediction, a dual-predictor consensus rule was applied: a candidate was excluded as an allergen only when AllerTOP v2.0 and AllergenFP v1.0 concordantly predicted an allergenic profile. For peptides under 16 amino acids (including MHC-I and MHC-II candidates) where AllergenFP is inapplicable due to length constraints, classification relied solely on AllerTOP v2.0. Epitopes flagged as allergenic were removed from further candidate selection (Sanami et al., 2023).

F. Human Self-Homology

To minimize the risk of autoimmune cross-reactivity and to exclude epitopes likely to be immunologically silent due to central tolerance, all candidate T-cell and B-cell epitopes surviving toxicity and allergenicity screening were additionally screened via BLASTP against the reviewed human reference proteome (UniProt Homo sapiens, Swiss-Prot subset) (Sharma et al., 2022). Epitopes meeting all of the following criteria were flagged as “self-like” and excluded from further consideration: percent identity $\geq 70\%$, alignment coverage $\geq 70\%$ (alignment

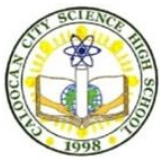
length / peptide length), and E-value $\leq 1e-5$, consistent with homology thresholds established for toxin-screening BLASTP (Sharma et al., 2022).

G. Population Coverage Analysis

To eliminate structural and immunological redundancy within the final chimeric construct, cross-class global sequence deduplication was applied prior to priority ranking. Candidate epitopes sharing an overlapping sequence frame or core subsequence with any previously selected candidate—regardless of target protein or MHC class—were sequentially excluded. This global frame-check ensured that each antigenic region was represented only once, maximizing sequence diversity and preventing redundant epitope coverage. The procedure for the HLA prediction for the computed population coverage analysis went as follows. MHC I binders were identified as peptides whose percentile rank $\leq 2\%$ (Shabbir et al., 2025). While MHC II binders were defined as peptides whose percentile rank was $\leq 10\%$ (Sanami et al., 2023). The binders were then translated to 2-field allele notation (e.g., HLA-A*02:01) to match allele-frequency tables downloaded from the Allele Frequency Net Database (AFND) (Gonzalez-Galarza et al., 2019). For each peptide and group of HLA alleles, their chromosome allele frequencies per locus were summed ($s = \sum \text{freq}[a]$, capped at 1.0), converted to per-locus coverage $= 1 - (1 - s)^2$, and combined loci assuming independence: $\text{overall_coverage} = 1 - \prod_{\text{loci}} (1 - \text{coverage_locus})$ (Shabbir et al., 2025). Peptides with overall_coverage $\geq 90\%$ were prioritized as population-covering candidates, (Ahmad et al., 2025). The population coverage was treated as a ranking criterion rather than a binary elimination filter. Candidates were ranked based on projected coverage metrics (e.g., IEDB Population Coverage Tool), allowing the selection of top-performing MHC-II peptides even when individual cumulative scores fell below 90% (Ahmad et al., 2025). Additionally, missing allele frequencies were treated with a score of 0 (Shabbir et al., 2025). The cumulative vaccine coverage was calculated via the union of all binding alleles.

H. Linker Integration and Adjuvant Attachment

To assemble the vaccine, linkers such as AAY, GP GPG, and KK would be utilized to connect Cytotoxic T-cells, Helper T-cells, and B-cells, respectively. Additionally, the inclusion of the Beta-defensin 3 to the N-terminus of the sequence using an EAAAK linker would also be



incorporated (Ahmad et al., 2024). Prior to final assembly, each selected epitope was screened to confirm the absence of embedded linker motifs (AAY, GPGPG, KK) within its native sequence, preventing ambiguity in linker-boundary parsing during downstream splicing operations (Section IV.A).

I. Experimental Corroboration

To verify that the peptides of the structure were observed experimentally, the final linked construct (Section I.G) was cross-referenced against IEDB's curated experimental record following construct finalization. For every construct epitope, the IEDB Epitope database was queried for records annotated to the source antigen (Mpox L1R, B5R, A35R; HIV-1 gp120, gp41, p24, p17) or their curated orthologs, restricted to entries carrying at least one positive assay outcome — MHC-ligand elution mass spectrometry, T-cell assay, or B-cell assay — rather than sequence-only or purely predicted annotations. Three corroboration tiers were assigned:

EXPERIMENTALLY CONFIRMED (exact sequence match to, or full containment within, a curated assay-backed IEDB epitope), **OVERLAP** (≥ 8 contiguous amino acids shared with a curated assay-backed epitope, without exact match or containment), and **NOVEL** (no IEDB record satisfying either criterion). Because the IEDB catalog is disproportionately weighted toward vaccinia virus relative to monkeypox virus — a legacy of decades of smallpox-vaccine research — and because A35R, B5R, and L1R possess direct vaccinia orthologs, matches to vaccinia-derived curated epitopes were retained as corroborating evidence but permanently flagged as cross-species (orthopoxvirus) rather than mpox-specific, consistent with the established mechanism by which orthopoxvirus cross-reactivity underlies smallpox-vaccine-conferred mpox protection. Cross species matches were not excluded.

To guard against false-positive corroboration arising from short-peptide coincidence within the ~200 kb poxvirus proteome, or from nomenclature mismatch between UniProt's orthopoxvirus gene identifiers (e.g., "Ankyrin repeat protein OPG189") and the construct's protein designations (e.g., "B5R"), every candidate IEDB match was independently realigned against the Phase I.A reference sequence set for its corresponding protein/variant. A match was retained only where pairwise identity to a Phase I.A reference fell within 83–100%; this check was applied uniformly to all candidate matches, and none was rejected on this basis. This step is

descriptive, not selective: no epitope was added, removed, reordered, or re-scored based on its corroboration outcome, and a NOVEL classification was not treated as disqualifying.

II. Structural Modelling and Verification

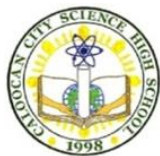
A. Secondary Screening of Post Construct Vaccine

a. Post Toxicity and Allergenicity Screening

For the screening of the whole vaccine sequence's toxicity and allergenicity, toxicity tests included sequence tests such as a Sliding-window hydrophobicity fraction, Cysteine check, ToxinPred2 1.1, Junctional peptides + MHC binding, Hemolysis, and BLASTP to UniProt Tox-Prot (Shabbir et al., 2025).

The hydrophobicity fraction test was conducted by first dividing a sequence into short windows of length 8, 12, and 15 amino acids (Sanami et al., 2023). For each window, hydrophobic residues were counted, and if any of the windows contained $> 80\%$ of hydrophobic residues, the construct was rejected (Waqas et al., 2023). The Cysteine check was performed for short junction peptides (8–20 aa) constructs that had > 2 cysteines were rejected from the list of candidates, adjuvants were exempted (Ahmad et al., 2025). ToxinPred2 1.1 was run throughout the entirety of the construct, and candidates were rejected after its prediction was concluded to be "toxin" with the accompaniment of a real BLAST hit or MERCI motif (Sharma et al., 2022). A junctional peptides analysis was conducted via MHC binding to screen the aforementioned linkers (Shabbir et al., 2025). Every linker was generated with all peptides spanning the junction of lengths of the following per junction: MHC I: 8–11 aa; MHC II: 12–20 aa (Waqas et al., 2023). A percentile rank of $\leq 2\%$ for MHC I and $\leq 10\%$ for MHC II, as well as the accompanying result of "toxin," which came from the ToxinPred2 1.1, highlighted the junctions as toxic and was rejected (Shabbir et al., 2025).

Post-assembly allergenicity screening included composition heuristics and sequence-based predictors such as Q/N Fraction, Charged Residue Count,



AllerTop, and AllergenFP (Sanami et al., 2023).

For each full construct and each junctional peptide, we computed $QN_Ratio = (\text{count}(Q) + \text{count}(N)) / \text{Length}$ and $\text{Surface_Charge} = \text{count}(D + E + H + K)$ (Shabbir et al., 2025). Junctional peptides were enumerated at every linker/boundary (MHC-I lengths 8–11 aa; MHC-II lengths 12–20 aa) and tested the same way as epitope candidates (Waqas et al., 2023). The following rules were applied: $QN_Ratio > 0.30$ flags the sequence as allergenic; Surface_Charge greater than 4 residues or 2% of the sequence length (rounded off) flags the sequence for review for peptide lengths greater than 20 aa (for short junction peptides, a raw $\text{Surface_Charge} > 4$ is automatically deprioritized) (Shabbir et al., 2025). All sequences were also submitted to at least two independent allergen predictors, AllerTOP and AllergenFP (Sanami et al., 2023). Final calls used a consensus rule: a full construct was rejected if 2 predictors classified it as allergenic, or if $QN_Ratio > 0.30$ combined with ≥ 1 predictor positive; a junctional peptide causing a positive prediction by ≥ 2 predictors caused construct rejection (Ahmad et al., 2025).

b. Solubility and Physicochemical Stability

The analysis of solubility as solubility is an important factor to be considered, since it dictates the effectiveness of reproduction through recombinant systems (Shabbir et al., 2025).

For each vaccine construct computation for their Length, Molecular_Weight (Da), Theoretical_pI, GRAVY (Kyte–Doolittle average), Aliphatic_Index, Instability_Index (ProtParam), and Estimated_HalfLife (ProtParam output), were conducted to verify their structural stability (Shabbir et al., 2025). Length, molecular weight, aliphatic index and estimated half-life were computed for reporting and experimental planning and were used as contextual flags only. Hard rejection decisions were based on direct solubility/stability predictors (Instability_index, GVI, DeepSol S2, CamSol) and theoretical pI value.

An Instability index of > 40 , a GRAVY score of > 0.4 , theoretical pI close to the

intended buffer pH of 7.0 (within ± 0.5 pH units) were all rejected (Savojardo et al., 2024). Additionally, the prediction of Global solubility was conducted using a machine-learning solubility predictor that utilizes a deep convolutional neural network (CNN), DeepSol S2, and an intrinsic solubility propensity predictor, CamSol (Shabbir et al., 2025; Ammar et al., 2026). Aggregation-prone regions reported by CamSol Intrinsic were scored and classified: < -1 poorly soluble, -1.0 to $+1.0$, moderately soluble. $> +1$ highly soluble (Ahmad et al., 2025). DeepSol S2 probability ≥ 0.5 was used to call a sequence “predicted soluble”, and a score of < 0.5 was called “predicted insoluble” (Ammar et al., 2026). Sequences with regions of high-scoring aggregation patches were rejected. Additionally, in accordance with the multi-tier selection logic, sequence flags required orthogonal structural corroboration: candidates were rejected only if a low DeepSol S2 score (< 0.50) coincided with a structure-aware aggregation patch spanning > 8 contiguous residues in a high-confidence domain (pLDDT ≥ 70).

B. Secondary and Tertiary Structure Prediction

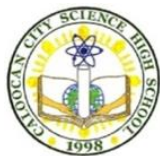
The study would utilize PsiPred and NetSurfP-3.0 for the secondary structure prediction of the multiepitope vaccine (Shabbir et al., 2025).

AlphaFold3 would be incorporated to construct a Tertiary (3D) Structure Prediction in order to create a 3D model of how the amino acids of the vaccine fold. Human Toll-Like Receptor (TLR) models, specifically TLR-2 (6NIG) and TLR-4 (8WTA), were taken from the RCSB Protein Data Bank (RCSB PDB) website (Krokidis et al., 2025 ; Fu et al., 2025).

C. Fold-Confidence Analysis

To evaluate the protein construct’s fold, the AlphaFold3’s fold confidence tool was utilized. For each AlphaFold3 job, all 5 ranked models were retained rather than only the top-ranked model. The following metrics were extracted directly from the AlphaFold3 output for each of the five models:

- pTM (predicted TM-score) — a global measure of predicted fold accuracy relative to a hypothetical true structure, used to flag models below the 0.5 threshold conventionally associated with a correctly predicted overall fold (Krokidis et al., 2025).
- PAE (Predicted Aligned Error) — the full residue-pair PAE matrix was retained (not



just its mean) to identify domain-level uncertainty: low intra-domain PAE combined with high interdomain PAE indicates confidently folded epitope blocks connected by an uncertain linker/hinge region, which is expected and distinguished from uniformly high PAE, which indicates the model is unreliable throughout.

- **fraction_disordered** — the proportion of residues AlphaFold3 assigns to a disordered rather than ordered state, used to flag constructs where a large fraction of the linked epitope chain is predicted to lack stable tertiary structure, since linker-heavy multi-epitope constructs are inherently prone to this failure mode.
- **ranking_score** — AlphaFold3's composite permodel score (incorporating pTM, ipTM where applicable, and disorder), used only to confirm the server's own model ranking; it was not used as a standalone pass/fail criterion given its composite nature.

In addition to the per-model confidence metrics above, the five ranked models were compared pairwise to test whether AlphaFold3 converged on a single fold or produced discordant alternative folds with superficially similar confidence scores. All C α atoms of each model pair were structurally aligned using Bio.PDB.Superimposer, and the pairwise C α root mean-square deviation (RMSD) was computed for all C (5,2) = 10 model pairs following superposition:

$$RMSD = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(r_i^{(A)} - r_i^{(B)} \right)^2}$$

Where $r_i^{(A)}$ and $r_i^{(B)}$ are the aligned C α coordinates of residue in models *A* and *B* after Superimposer's rotation/translation fit, and *N* is the number of C α atoms compared. The resulting 5×5 pairwise RMSD matrix was used to flag high inter-model divergence (RMSD substantially elevated relative to the within domain baseline) as evidence of an unstable or ambiguous predicted fold, independent of any single model's individual pTM or ranking_score.

A model was accepted if it satisfied all of the following: pTM $\geq$ 0.5; fraction_disordered below a construct-appropriate threshold accounting for expected linker flexibility;

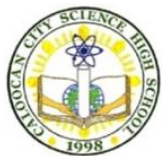
PAE indicating low intradomain uncertainty (permitting elevated interdomain/linker PAE); and pairwise C α RMSD against the other four ranked models consistent with convergent, rather than discordant, folding.

D. Solubility Analysis

Each structural model was converted into the Protein Data Bank format (.pdb) and was prepared in the following manner:

- Repaired/minimally relaxed if the server provided that option
- Protonated at pH 7.0
- Analyzed for solvent accessible surface area (FreeSASA 2.2.1) and hydrophobic surface fraction
- Scored for structure-aware aggregation propensity (Aggrescan3D and CamSol structural mode)
- Evaluated for electrostatic surface potential (PDB2PQR and APBS)
- Checked for predicted overall solubility from sequence (DeepSol S2).

Per-residue model confidence (pLDDT) was recorded and only aggregation/patch calls in regions with pLDDT $\geq$ 70 were treated as high-confidence (Jumper et al., 2021). Decision rules combined sequence and structure evidence: constructs were flagged for rejection when DeepSol S2 predicted insolubility (probability $<$ 0.50) and a structure-aware tool (Aggrescan3D) identified an exposed, high-scoring aggregation patch $>$ 8 contiguous residues in a pLDDT $\geq$ 70 region (Badaczewska-Dawid et al., 2022; Ammar et al., 2026; Pimta Wong et al., 2025; Planas-Iglesias et al., 2024). Additional structural flags included hydrophobic surface fraction $>$ 0.25 or exposed hydrophobic patch area $>$ 250 Å². These were evaluated as quantitative physical signatures of intrinsic disorder across extended linker cassettes rather than standalone rejection criteria. Candidate surface residue interactions were assessed for overall thermodynamic stability using FoldX 5 total folding energy (FoldX_Total_Energy_kcal_mol). Constructs exhibiting anomalously high positive folding energies were flagged for potential structural compromise and removed from candidate selection. (Mitternacht, 2016; Hebditch et al., 2017; Schymkowitz et al., 2005; Baker et al., 2001). All repaired PDBs, per-residue tables (pLDDT, FreeSASA 2.2.1, hydrophobic_FreeSASA 2.2.1, Aggrescan3D/CamSol scores), DeepSol S2 outputs, APBS maps, and the final PASS/REVIEW/REJECT decision with a short reason code were archived as Supplementary Data.



E. Structure Minimization and Model Validation

Structure minimization and repair were performed by the Phenix Command Line to adjust the atomic coordinates of the protein model (Chundru & Ghanem, 2026). The utilization of the Ramachandran Plot was conducted and generated using MolProbity, with the preferred quality being 95-98% of residues within favorable conditions (Qiao et al., 2026). A MolProbity score of 0.5-1.5 was deemed excellent quality, a score < 2 was acceptable quality, and a score of > 3 was determined as poor quality (Qiao et al., 2026). This was conducted to verify the stereochemical quality of the protein backbone. By ensuring that the torsion angles (ϕ and ψ) fall within the allowable conformational spaces (Shabbir et al., 2025).

III. Interaction and Simulation

A. Receptor and Ligand Preparation

Prior to docking, both receptor structures were inspected against their raw RCSB depositions to identify non-native chain content and occupied binding pockets that would otherwise corrupt the docking run if the depositions were used unmodified.

The deposited TLR-2 (6NIG) structure is a crystallization chimera: the modeled "TLR2" chain is not full-length human TLR2 alone but a fusion of the human TLR2 ectodomain with a variable lymphocyte receptor B (VLR-B) scaffold from the hagfish *Eptatretus stoutii*, incorporated to facilitate crystallization. Per the deposition's _struct_ref_seq records, author-numbered residues 1–507 correspond to human TLR2 (UniProt O60603). Residues 509–576 correspond to VLR-B (UniProt Q2YE02), and residue 508 is an engineered linker joining the two. Docking the unstripped chain would allow the vaccine construct to dock against the non-human VLR-B scaffold and be scored as though it were a human TLR2-binding event. The VLR-B segment (residues 508–576) was therefore excised, and only human TLR2 ectodomain residues 27–507 (478 residues; residues 1–26 corresponding to the unmodeled native signal peptide) were retained for docking.

Additionally, 6NIG was deposited in complex with ligand KQD (Diprovocim), a synthetic small-molecule TLR2 agonist occupying the receptor's ligand-binding pocket. Because this is the same site the vaccine construct's β -defensin 3/TLR2-engaging epitope surface is intended to probe, Diprovocim was removed prior to docking to avoid steric and energetic exclusion of the vaccine construct from its own target pocket by a co-crystallized small molecule.

The TLR-4 (8WTA) was fully retained, TLR-4 chain A (residues 27–627) and MD-2 chain C (residues 19–158) were kept together as a single 741-residue receptor complex.

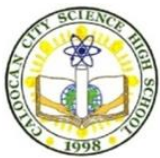
The final prepared inputs for Molecular Docking were: TLR2 chain A, residues 27–507 (478 residues, VLR-B and Diprovocim removed); and TLR4/MD-2, chains A+C (TLR4 residues 27–627 plus MD-2 residues 19–158, 741 residues total, lipid-A acyl chains retained). Preparation logs, including residue-range justification and removed HETATM/chain records, were archived as Supplementary Data.

B. Molecular Docking

To evaluate innate immune receptor activation, molecular docking against TLR-2 and TLR-4 was performed using the high-confidence adjuvant domain (residues 1–45; human β -defensin 3). In structural predictions (AlphaFold3), the downstream epitope cassette (residues 51–566) exhibited low per-residue confidence (pLDDT < 50) and high inter-model RMSD (18.5 - 23.9 Å), reflecting the intrinsic flexibility imparted by the GPGPG, AAY, and KK linkers designed for proteasomal processing rather than a rigid global tertiary fold. Because TLR-1/2 and TLR-4 agonism is driven specifically by the β -defensin 3 adjuvant moiety, docking and subsequent molecular dynamics (MD) simulations were restricted to the adjuvant–receptor interface to ensure physical reproducibility and structural rigor.

In order to perform Molecular Docking, the study utilized Haddock v2.4 by taking the AlphaFold3 model of the construct's high-confidence adjuvant domain and TLR-2 and TLR-4 receptors and docking it with the 3-dimensional construct. The lowest binding energy (kcal/mol) is considered an optimal fit for each of the receptors. Results would be based on the Global Minimum Energy or the lowest energy structure within the most populated cluster (Shabbir et al., 2025).

Additionally, to more accurately represent the physiologically relevant receptor architecture, docking was performed against TLR-4 in complex with its obligate co-receptor MD-2, rather than the TLR-4 ectodomain in isolation, given that TLR-4 signaling requires MD-2 for ligand engagement and downstream activation (Fu et al., 2025). The TLR-4/MD-2 complex structure was retrieved from RCSB PDB (8WTA), which includes the co-receptor in its deposited coordinates. For TLR-2, which was modeled as a homodimer-competent ectodomain (6NIG) without a specific heterodimeric partner, docking results are interpreted as representative of the TLR-2 ligand-binding groove rather than a specific TLR2/TLR1 or TLR2/TLR6 heterodimeric



signaling complex; this is noted as a limitation, since differential TLR2 heterodimerization can influence downstream signaling specificity (Rcheulishvili et al., 2023).

C. 100ns MD Simulation

The study utilized a High-Performance Computing (HPC) cluster running GROMACS to evaluate the dynamic stability of the vaccine-receptor complex and to perform a continuous 100 ns simulation. The 100 ns timeframe is regarded as sufficient for reaching equilibrium in molecular simulations and is the most commonly used standard (Halder et al., 2025). This was conducted in order to get the metrics of Root Mean Square Deviation (RMSD) and Root Mean Square Fluctuation (RMSF) over 100ns. The graph of RMSD with plateaus below 3.0 Å will serve as proof of stable interaction. In complement with the RMSD graph, RMSF peaks are examined in order to identify the structural flexibility of the epitopes, in order to ensure they remain accessible for immune recognition. Additionally, the Ratio of Gyration (R_g) will also be observed. A stable R_g is represented by a flat line, which indicates high stereochemical quality; an increasing R_g is represented by an upward slope, which dictates that the protein is unfolding. Lastly, a decreasing R_g is represented by a downward slope, which states that the loosely folded protein is collapsing into a more compact shape.

D. Structural Flexibility and Deformability

The calculation of eigenvalues and covariance maps, deformability, and B-factors helps predict the protein-protein complex's molecular movements and structural dynamics in a biological environment. A higher eigenvalue implies a more stable and viable rigid vaccine construct, covariance maps where red represents a positive correlation (residues moving toward the same direction), blue represents an anti-correlation (residues moving opposite of each other), and white, which stands for no correlation (random or independent direction), deformability results were interpreted in the form of graphs, where high peaks indicated flexible regions while low valleys concluded rigid regions. Lastly, B-factor metrics were calculated by multiplying the RMSD value by $8\pi^2$

$$(B_i = \frac{8\pi^2}{3} (\Delta R_i^2)) \text{ (Shah et al., 2025).}$$

E. mRNA Construct Design, Plasmid Templating, and LNP Formulation

a. Plasmid Template Assembly and In Silico Cloning

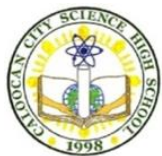
To transition the dual-pathogen chimeric construct into a non-viral, highly expressible synthetic mRNA platform, the candidate genetic sequence was adapted into a linear eukaryotic mRNA transcript topology consisting of a Cap-1 5' terminus, a human -globin (H B B) 5' untranslated region (UTR), an

optimized Kozak consensus sequence (5'-GCCACCATG-3'), the original chimeric structure itself (open reading frame- ORF), a dual human -globin / AES 3' UTR, and a segmented 130-nucleotide polyadenylation tail (A130).

Table 3.
mRNA Transcript Architecture

Linear mRNA Transcript Architecture						
Cap-1 AG	5' UTR	Kozak	Signal Peptide (tPA)	β -defensin 3-EAAAK- Chimeric ORF	3' UTR	Poly(A)130

To optimize eukaryotic translation efficiency and transcript longevity, the synthetic construct was organized into a linear eukaryotic mRNA architecture comprising six essential functional elements. At the 5' terminus, a CleanCap® AG co-transcriptional analog installs an endogenous Cap-1 structure (m7G(5')ppp(5') (2'-O-methylA)pG) to prevent premature exonucleolytic degradation and evade IFIT1-mediated cellular immune recognition with greater than 95% efficiency (Henderson et al., 2021). Flanking the coding region, a human untranslated region (UTR) facilitates unhindered assembly of the 40S ribosomal pre-initiation complex without forming stable secondary hairpins (Leppek et al., 2022). Translation initiation is directed by a canonical Kozak consensus signal (5'-GCCACCATG-3') placed immediately upstream of the start codon to optimize ribosome loading (Narožna et al., 2021). The central chimeric open reading frame (ORF) was codon-adapted to align with human tRNA abundance (CAI>0.85; 50-60% GC content) and synthesized with 100% N1-methylpseudouridine (N1 Ψ TP) substitution to suppress intracellular TLR3/7/8 sensing (Joshi et al., 2023; Mulrone et al., 2024). Topological evaluation was performed using DeepTMHMM to confirm the absence of unintended hydrophobic transmembrane domains, ensuring optimal cytosolic solubility and protein processing (Hallgren et al., 2022). To favor extracellular exposure of the B-cell epitope arm and surface-level engagement of the β -defensin 3 adjuvant with TLR2/4 on antigen-presenting cells, a secretory signal peptide was appended to the true N-terminus of the construct, positioned upstream of the β -defensin 3-EAAAK adjuvant cassette. The human tissue plasminogen activator (tPA) signal sequence was selected for this role, as it is a well-characterized secretory leader routinely used in multi-epitope vaccine constructs to direct co-translational translocation into the endoplasmic reticulum without requiring pathogen-derived signal sequences (Naveed et al., 2022; Rcheulishvili et al., 2023). This routes the translated antigen through the secretory pathway, enabling extracellular or cell-surface antigen presentation



supportive of humoral (B-cell) and MHC-II–restricted helper T-cell responses, while retaining the potential for dendritic cell cross-presentation to sustain CD8+ cytotoxic T-lymphocyte priming.

Signal peptide cleavage was predicted and validated using SignalP 6.0 to confirm a high-confidence cleavage site immediately downstream of the tPA leader and upstream of the mature β -defensin 3 N-terminus, ensuring that the adjuvant domain is not truncated or disrupted by aberrant cleavage-site placement (Teufel et al., 2022). Constructs in which the predicted cleavage site fell within 3 residues of the intended boundary, or in which cleavage probability at the intended site fell below 0.80, were flagged for re-evaluation of linker spacing between the signal peptide and β -defensin 3 domain. Following signal peptide integration, the full construct was re-submitted to DeepTMHMM to confirm that the addition of the tPA leader did not introduce unintended internal transmembrane segments elsewhere in the sequence, preserving the topological integrity previously established for the unmodified construct.

Downstream, a dual human γ -globin / AES 3' UTR stabilizes the transcript against cytosolic deadenylation (Leppek et al., 2022), while a segmented 130-nucleotide polyadenylation tail (A100-linker(L10)-A20 ;L10=5'-GCATATGACT-3') binds Poly(A)-Binding Proteins (PABPs) to maximize intracellular half-life and mitigate recombinational loss during bacterial plasmid propagation (Grier et al., 2020; Wei et al., 2023). Secondary structure minimum free energy (G) was computed using RNAfold (ViennaRNA Package) to confirm that the 5' UTR and Kozak region avoided stable hairpin loops that impede ribosome scanning (Lorenz et al., 2021).

1. Microfluidic Lipid Nanoparticle (LNP) Encapsulation

In silico cloning was simulated in SnapGene software to construct an in vitro transcription (IVT) template within a high-copy artificially engineered plasmid vector (pUC57) backbone. The mRNA cassette was positioned under the regulatory control of a bacteriophage T7 promoter sequence (5'-TAATACGACTCACTATAGGG-3') to enable high-yield T7 RNA polymerase transcription (Mendoza et al., 2022). Directional cloning was executed by introducing a 5' *Bam*HI restriction site upstream of the 5' UTR and a 3' *Hind*III site downstream of the poly(A) tail. To ensure precise run-off transcription without transcribing extraneous plasmid backbone sequences, a Type IIS restriction site (BspQI) was engineered directly adjacent to the terminal adenine of the A130 tail (Gaucher et al., 2023). Reading frame orientation, promoter alignment, and cassette integrity were validated via

simulated restriction digestion and in silico PCR amplification (Shabbir et al., 2025)

Plasmid Template Assembly and In Silico Cloning

To avoid triggering endosomal pattern recognition receptors (TLR3, TLR7, TLR8) and RIG-I cytosolic sensors that trigger transcript degradation and premature translation arrest, 100% substitution of uridine triphosphate (UTP) with N1-methylpseudouridine triphosphate (N1 Ψ TP) was modeled (Morais et al., 2021; Mulrone et al., 2024). Co-transcriptional capping using CleanCap® AG reagent was integrated into the T7 IVT pipeline, yielding an endogenous Cap-1 structure that evades IFIT1-mediated translation repression with >95% efficiency (Henderson et al., 2021). Enzymatic plasmid template removal via DNase I digestion and single-step linearized mRNA recovery were computationally integrated.

3.4.4 Microfluidic Lipid Nanoparticle (LNP) Encapsulation

The mRNA sequence was encapsulated within a four-component lipid nanoparticle matrix using the benchmark SM-102 formulation at a 50 : 10: 38.5: 1.5 molar ratio (Hou et al., 2021; Schoenmaker et al., 2021):

- i. Ionizable Cationic Lipid: SM-102 (50 mol%), facilitating pH-dependent electrostatic mRNA complexation and cytosolic endosomal escape (Hassett et al., 2021)
- ii. Helper Phospholipid: 1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC; 10 mol%), conferring structural bilayer integrity (Cheng & Lee, 2020).
- iii. Sterol: Synthetic cholesterol (38.5 mol%), regulating membrane fluidity and nanoparticle stability (Eygeris et al., 2022).
- iv. PEGylated Lipid: 1,2-dimyristoyl-rac-glycero-3-methoxypolyethylene glycol-2000 (DMG-PEG2000; 1.5 mol%), preventing particle aggregation and extending circulation half-life (Tenchoy et al., 2021).

Microfluidic advection mixing via a staggered herringbone mixer (SHM) was modeled at a 3:1 aqueous (100 mM sodium acetate, pH 5.0) to organic (absolute ethanol) flow rate ratio (FRR) at an amine-to-phosphate (N/P) ratio of 6.0 (Rosas-Hernandez et al., 2023). Subsequent neutral pH dialysis (pH 7.4) was modeled to yield uniform mRNA-LNPs (80-100 nm, PDI < 0.2) optimized for dendritic cell endocytosis and cytosolic expression.

Probabilistic Immune Simulation and Population Profiling



To model system-level immune kinetics accurate to the target demographic, probabilistic immune simulations were conducted using the C-ImmSim server over a three-dose regimen administered at days 0, 28, and 56, simulated across 1,000 time steps, wherein each step corresponds to 8 hours (Shabbir et al., 2025). C-ImmSim employs a cellular automaton-based immune simulator that utilizes Position-Specific Scoring Matrices (PSSM) to model peptide-HLA binding, systemic immune cell interactions, clonal proliferation, and cytokine secretion (Rapin et al., 2010).

Because C-ImmSim natively models antigen as instantaneously available at the moment of dosing, an adjustment was required to reflect the platform-specific pharmacokinetics of mRNA-LNP delivery, wherein antigen presentation is contingent on LNP endosomal escape and subsequent host-cell translation rather than immediate availability (Hassett et al., 2021). To approximate this, each simulated dose (days 0, 28, and 56) was modeled as a delayed-onset antigen pulse rather than a bolus input: antigen availability was set to begin at approximately 6–8 hours post-dose, consistent with reported onset of detectable protein expression following LNP cellular uptake and endosomal escape (Hassett et al., 2021; Schoenmaker et al., 2021), and to ramp toward peak availability over the subsequent 24–48 hours before gradual decay, approximating the transient expression profile typical of mRNA constructs. This staged-input approach was applied uniformly across all three doses to preserve internal consistency across the booster series.

To account for the local immunological landscape of the Southeast Asian (Austronesian) proxy coverage, host HLA profiles were parameterized using the dominant regional MHC Class I (HLA-A*24:02, HLA-B*15:02, HLA-B*40:01) and MHC Class II (HLA-DRB1*15:01, HLA-DRB1*12:02) allele frequencies drawn from the same IEDB allele-frequency tables used in the Population Coverage Analysis (Section II.C.I.F) (Waqas et al., 2023). A secondary heterozygous profile and a default control profile were simulated alongside the dominant local profile to evaluate intra-population variance in immune response. T-cell activation was tracked through active CD4⁺ Helper and CD8⁺ Cytotoxic T-lymphocyte counts and cytokine production profiles (IFN- γ , IL-2), with clonal diversity of the responding T-cell repertoire assessed via the Simpson diversity index (D) generated natively by C-ImmSim, wherein a decreasing D over the simulation window indicates the emergence of dominant, epitope-specific T-cell clones (Rapin et al., 2010).

B-cell dynamics were concurrently monitored through subpopulation differentiation (active, plasma, and memory B-cells) and antibody isotype class switching from IgM to high-affinity IgG1 and IgG2 following each booster administration. Consistent with the study's evaluation criteria, success was interpreted as a significant post-boost increase in IgG titers and IFN- γ production relative to baseline (Shabbir et al., 2025).

As C-ImmSim does not provide a native module for modeling intracellular mRNA translation kinetics, the delayed-onset pulse described above represents an approximation rather than a mechanistic simulation of LNP uptake and endosomal escape; this simplification is noted as a limitation of the immune simulation stage.

IV. Immunodominance and Interference Analysis

A. Sequence Deconstruction and Splicing of Individual Pathogen Component

1. Splicing Procedure

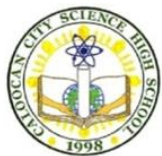
The finalized post-linker, post-adjuvant MpoxHIV sequence produced at the end of Section I.G (Linker Integration and Adjuvant Attachment) served as the input for splicing. Using the epitope-origin annotations retained from Sections I.C–I.F, each epitope block in the assembled sequence was tagged by source pathogen (HIV-1 CRF01_AE vs. Mpox Clade IIa/IIb) prior to segmentation. Evaluating these components independently is a critical validation step for multi-epitope vaccines targeting co-infections to ensure neither pathogen's antigens are structurally or immunologically compromised (Al-Madhagi, 2024; Li et al., 2026).

Two daughter constructs were generated via Biopython (SeqRecord) sequence slicing: Construct-HIV, comprising all HIV-1-derived CTL, HTL, and B-cell epitopes and retaining their original AAY (CTL–CTL), GPGPG (HTL–HTL), and KK (B-cell–B-cell) linkers in their original relative order, and Construct-Mpox, comprising all Mpox-derived CTL, HTL, and B-cell epitopes under the same linker scheme.

The β -defensin 3 adjuvant was retained at the N-terminus of both daughter constructs via the same EAAAK linker used in the parent construct (Ahmad et al., 2024), so that adjuvant contribution is held constant across both components and cannot itself explain any difference observed in Section C.

2. Post-Splice Secondary Re-Screening

Splicing creates at least one new junction that did not exist in the parent construct, the point where two epitope blocks formerly separated by the excised pathogen's sequence become directly adjacent. Because this junction, and the altered local composition surrounding it, were never evaluated during Phase I or Phase II, each daughter construct was



independently resubmitted to the full Section I.E–I.F and Section II.A screening cascade before proceeding to structural modeling.

Antigenicity was re-scored under the Kolaskar & Tongaonkar propensity scale (mean threshold ≥ 1.00 , 7-residue sliding window) and ESM-2 (threshold ≥ 0.50), with consensus agreement required as in Section I.C. The novel splice-junction peptide was additionally screened using the same junctional-peptide MHC-binding rule from Section II.A.a (MHC I: 8–11 aa, percentile rank $\leq 2\%$; MHC II: 12–20 aa, percentile rank $\leq 10\%$) combined with a ToxinPred2 1.1 “toxin” call, with a positive result under both criteria triggering rejection of the daughter construct.

The whole construct was re-screened for toxicity via sliding-window hydrophobicity fraction (8/12/15-aa windows, $> 80\%$ reject), Cysteine count (> 2 in an 8–20 aa junction, reject; exempted for adjuvants), ToxinPred2 1.1, HemoPI2 1.3, and BLASTP against UniProt Tox-Prot, using the identical thresholds defined in Sections I.E and II.A.a. Re-screening is necessary because removing roughly half the sequence changes local composition around every remaining window, not only the new junction. Allergenicity was likewise re-screened via Q/N Fraction (> 0.30 flag), Charged Residue Count (> 4 flag), AllerTop, and AllergenFP, applying the same two-predictor consensus rejection rule used in Section II.A.a.

Physicochemical stability: Length, Molecular Weight, Theoretical pI, GRAVY, Aliphatic Index, Instability Index, and Estimated Half-Life, was recomputed via Bio.SeqUtils.ProtParam for each daughter construct, applying the same hard-rejection thresholds from Section II.A.b (Instability Index > 40 , GRAVY > 0.4 , pI within ± 0.5 pH units of 7.0) and the same DeepSol S2/CamSol solubility calls. Instability Index and pI are dipeptide-composition- and net-charge-dependent respectively, so a daughter construct is not guaranteed to pass simply because it was assembled from epitopes that individually passed as part of the whole.

Only daughter constructs that pass all criteria above are carried forward to the next step. A construct that fails is treated as

a limitation requiring adjustment of the splice boundary (e.g., re-ordering which epitope sits adjacent to the cut).

B. Independent Structural Modeling and Immunogenic Validation (Phase II–III Replication)

Construct-HIV and Construct-Mpox were each carried through the identical Phase II and Phase III pipeline in parallel, independent runs, using the same tools and acceptance thresholds applied to the original chimeric construct.

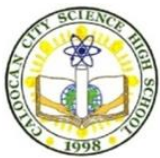
1. Structural Modeling and Verification

Secondary structure was predicted via PsiPred and NetSurfP-3.0, and tertiary structure via AlphaFold3, generating one 3D model per daughter construct. Solubility analysis was conducted via FreeSASA 2.2.1 and hydrophobic surface fraction, Aggrescan3D and CamSol structural mode, PDB2PQR/APBS electrostatic surface potential, and DeepSol S2, applying the same pLDDT ≥ 70 high-confidence threshold and rejection rules from Section II.C. Model validation was performed via MolProbity Ramachandran analysis (target: 95–98% residues in favored regions; MolProbity score 0.5–1.5 = excellent), followed by Phenix Command Line minimization and repair (Yin et al., 2022).

2. Interaction and Simulation

Molecular docking was performed by taking each daughter construct's AlphaFold3 model and docking it independently against TLR-2 (PDB: 6NIG) and TLR-4 (PDB: 8WTA) using Haddock v2.4, reporting the Global Minimum Energy structure for each complex (Sinani et al., 2022). Each docked complex was then run independently on the DOST-COARE HPC cluster via GROMACS for a continuous 100 ns MD simulation, with RMSD, RMSF, and Radius of Gyration extracted via MDAnalysis, applying the same stability criteria (RMSD plateau below 3.0 Å; flat Rg trace) (Mamun et al., 2025) used in Section III.B. Structural flexibility and deformability were assessed through eigenvalue, covariance map, deformability, and B-factor analysis via ProDy for each construct–receptor complex, as in Section III.C (Zhang et al., 2021).

Codon optimization and transcript adaptation were conducted by recalculating CAI (target > 0.85) and GC content (target 50–60%) per construct (Ahmed et al., 2023). Each daughter construct was in silico adapted into a linear eukaryotic mRNA transcript topology (Cap-1 5' terminus, human β -globin 5' UTR, GCCACCATG Kozak sequence, construct ORF, dual β -globin/AES 3' UTR, 130-nt segmented poly(A) tail, 100% N1-



methylpseudouridine substitution) (Hooper et al., 2024; Nance & Meier, 2021) exactly as established in Section III.D.

Finally, probabilistic immune simulation was conducted by simulating each mRNA construct independently — in isolation, with no antigen from the other pathogen present, in C-ImmSim over the same three-dose regimen (days 0, 28, and 56; 1,000 time steps of 8 hours each) (Khan et al., 2025), using the dominant Philippine HLA profile (Villanueva et al., 2020) (HLA-A*24:02, HLA-B*15:02, HLA-B*40:01; HLA-DRB1*15:01, HLA-DRB1*12:02) plus the secondary heterozygous and default control profiles established in Section III.E. Outputs recorded per construct included CD4⁺/CD8⁺ T-cell activation counts, IFN- γ and IL-2 production, the Simpson diversity index (D), B-cell subpopulation dynamics, and IgM $\rightarrow$ IgG1/IgG2 class-switching kinetics to assess how antigen presentation shapes the B-cell clonal hierarchy (Weidner et al., 2026).

C. Comparative Immunodominance Analysis

To generate a distribution suitable for inferential comparison, each construct (chimeric MpoxHIV, Construct-HIV, Construct-Mpox) was independently simulated in C-ImmSim $n = 10$ times per HLA profile, using a different random seed for each run (Cheng et al., 2023) while holding all antigen and dosing parameters constant. Peak IgG titer, time-to-peak, AUC, CD8⁺ activation, and Simpson diversity index (D) were extracted from each run to construct per-condition distributions for the t-test/Mann–Whitney comparisons.

1. Individual-vs-Individual Proportionality

Peak IgG titer, time-to-peak, AUC of antibody and cytokine curves, CD8⁺ CTL activation, and Simpson diversity index (D) were compared between the standalone Construct-HIV and Construct-Mpox outputs across all three simulated HLA profiles, using an independent samples t-test (or Mann–Whitney U where normality could not be assumed), $\alpha = 0.05$ (Fan et al., 2024). Responses were classified as proportional if no statistically significant difference was found.

4.3.2 Individual-vs-Combined (Overshadowing Test)

Each standalone construct's immune profile from Section B was compared against that same pathogen's contribution within the original chimeric MpoxHIV construct's Phase III C-ImmSim output (Section III.E). A statistically significant reduction in a pathogen's IgG titer, IFN- γ production, or CD8⁺ activation in the combined construct relative to its standalone counterpart was interpreted as

evidence of immunodominance, a phenomenon where dominant antigens actively repress or reprogram the immune hierarchy against subdominant epitopes (Li et al., 2026; Wan et al., 2026).

RESULTS AND DISCUSSION

Phase I: Epitope Selection and Construction

1.1 Sequence Retrieval

Retrieval through the NCBI Entrez API returned **202 non-redundant sequences across the seven target antigens**, against a design target of 210 (30 per antigen). HIV-1 gp120, gp41, p17 and p24 and Mpox L1R each returned the full 30. Mpox A35R returned 27 and Mpox B5R returned 25, because NCBI holds no further non-redundant Clade II records for these two genes. This is a limit of the public database rather than an artefact of the filtration pipeline.

A related constraint applied to clade annotation. The originally specified filter, which required an explicit "Clade IIa" or "Clade IIb" string in the record, returned **zero** B5R and A35R sequences: NCBI records for these genes frequently carry no clade metadata at any resolution. Sequences were therefore tagged *clade_confirmed* or *clade_unconfirmed* and both categories retained, an approach consistent with documented incompleteness in viral sequence annotation (Schäffer et al., 2020).

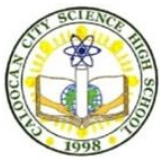
Table 4.

Sequence retrieval by antigen

Antigen	Pathogen	Retrieved	Target
gp120	HIV-1 CRF01_AE	30	30
gp41	HIV-1 CRF01_AE	30	30
p17	HIV-1 CRF01_AE	30	30
p24	HIV-1 CRF01_AE	30	30
L1R	MPXV Clade II	30	30
A35R	MPXV Clade II	27	30
B5R	MPXV Clade II	25	30
Total		202	210

1.2 Physicochemical Pre-screening of Source Antigens Of the 202 retrieved sequences, **35 were excluded for containing non-standard residues** (X, B, Z, J, U, gaps or stop codons), leaving **167 carried forward**. Of these, 75 returned an instability index below 40 (STABLE) and 92 returned a value at or above 40 (UNSTABLE_FLAGGED).

The instability index was recorded as an advisory flag rather than applied as an exclusion filter. All 27 A35R variants scored **41.03** — marginally above the threshold of 40 — and applying the criterion as a hard filter would have silently removed an entire required antigen from the design. Because the study's unit of selection is the short epitope rather than the source protein, and because the assembled construct receives its own independent physicochemical assessment at Step 2A (Section



2.2), source-protein instability is reported here for context only.

1.3 Antigenicity Screening

All **167 sequences** were classified **ANTIGENIC**, with **zero disagreements** between the Kolaskar–Tongaonkar propensity scale and the fine-tuned ESM-2 classifier.

A 100% pass rate means this step discriminated nothing at the source-protein level. This is the expected outcome: all seven target antigens are established immunogens with prior vaccine literature. The step is therefore reported as a **consistency check between two independent methods** — one propensity-based, one a protein language model — and not as a filter that reduced the candidate pool.

The ESM-2 classifier was fine-tuned on a curated set of **774 real sequences (390 antigenic from IEDB, 384 non-antigenic housekeeping proteins)**, achieving **98.6% held-out accuracy**. The methodology section's figure of 500 + 500 describes the original design target; the smaller set reflects a decision to use only genuine curated sequences rather than generate synthetic entries to reach a round number.

1.4 Epitope Prediction and Percentile Filtration Sliding-window k-mer generation across the 167 sequences produced **311,338 candidate epitopes** (MHC-I 9–10-mers, MHC-II 15-mers, B-cell 16-mers). Binding prediction used the IEDB MHC Tools API, with MHCflurry 2.2.1 and MHCnuggets 2.4.1 run locally for the coverage recomputation; B-cell linear epitopes were scored with BepiPred-2.0.

Percentile-rank filtration (MHC-I $\leq 1.0\%$, MHC-II $\leq 10.0\%$, BepiPred tiering as specified in Section II.C.I.D) retained **38,786 candidates**.

Table 5.

Phase I selection funnel

Step	Operation	Retained
1A	NCBI retrieval	202 sequences
1B	Non-standard residue removal	167 sequences
1C	Antigenicity consensus (K&T + ESM-2)	167 (100% ANTIGENIC)
1Da	k-mer generation and binding prediction	311,338 epitopes
1Db	Percentile-rank filtration	38,786
1Dc	Conservancy $\geq 20\%$	4,140
1Ea	Top-10 per antigen per class → toxicity screen	210 (all NON-TOXIC)
1Eb	Allergenicity consensus screen	139 (NON-ALLERGEN)

1Ec	Human self-homology (BLASTP vs Swiss-Prot)	139 (all NONE)
1F	Population coverage scoring	139 scored
1G	Construct assembly	32 epitopes → 566 aa

Note. Class composition after Step 1Db was B-cell 18,805, MHC-II 14,445 and MHC-I 5,536.

1.5 Conservancy Benchmarking

Conservancy analysis against the retrieved variant panels returned 4,140 epitopes at $\geq 20\%$ conservancy (the minimal benchmark), 1,922 at $\geq 50\%$ (the highly acceptable benchmark) and 954 at 100% across all source variants.

1.6 MHC-I Immunogenicity Ranking

BigMHC IM was applied to the surviving MHC-I pool, scoring 4,978 unique allele–peptide pairs [VERIFY — reconcile with the 5,536 MHC-I candidates entering this step]. 408 pairs were PRIORITIZED and 5,128 DEPRIORITIZED. BigMHC IM emits a bounded $[0, 1]$ sigmoid probability; the tool's own neutral midpoint of 0.50 was adopted as the prioritization boundary. Consistent with the methodology's wording, no candidate was excluded on immunogenicity score alone — the output functions as a ranking signal only.

1.7 Conformational B-cell Epitope Screening

This step did not execute. SEMA 2.0 documents a public REST API, but its prediction endpoints sit behind a commercial anti-bot system (ServicePipe). Re-verified endpoint by endpoint on 28 August 2026: the OpenAPI schema is served, but POST /proteins and all /proteins/{id}/*_data GET endpoints return a JavaScript challenge page even under a full browser User-Agent with browser headers, a warmed session and cookie replay. All seven antigens are recorded UNSCREENED.

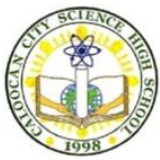
This is an annotation and completeness gap with no effect on the shipped sequence. Every B-cell slot in the final construct was determined by cross-class frame deduplication (Section 3.1.10) rather than by priority ranking, so no SEMA score could have arbitrated any selection actually made.

1.8 Toxicity, Allergenicity, and Self-Homology

Following the top-10-per-antigen-per-class shortlist (7 antigens $\times$ 3 classes $\times$ 10 = **210 candidates**), safety screening returned:

1.8.1 Toxicity

210/210 NON-TOXIC. Zero candidates produced a genuine BLASTP hit against the local UniProt Tox-Prot database or carried a MERCI toxin motif. Classification required positive evidence — a real homology hit or motif match — rather than a raw ToxinPred2 probability. Applied as a bare threshold, the raw score classified human serum albumin, human GAPDH, an unmodified linker sequence and the study's own β -defensin-



3 adjuvant as toxic, a false-positive mode the tool's authors themselves document (Sharma et al., 2022).

1.8.2 Allergenicity

210 → 139 NON-ALLERGENS. Exclusion required agreement between AllerTOP v2.0 and AllergenFP v1.0. Measured on the real candidate pool, the two predictors disagreed on **41 of the 70 peptides where both could run (59%)** and agreed on ALLERGEN for only 14. Treating either predictor as an independent exclusion signal would therefore have removed peptides on close to a coin flip. Where only one predictor applies — AllergenFP cannot process peptides below 16 residues, which covers every MHC-I and MHC-II candidate here — that predictor decides alone. A QN_Ratio above 0.30 continued to exclude unconditionally.

1.8.3 Human self-homology

139/139 returned Self_Homology_Status = NONE against the reviewed human Swiss-Prot proteome (20,431 sequences), indicating no molecular-mimicry or autoimmune liability anywhere in the surviving pool.

1.9 Population Coverage

1.9.1 Per-epitope coverage

Of the 139 screened candidates, **5 MHC-I and 12 MHC-II epitopes reached the $\geq 90\%$ prioritization criterion.** Consistent with the methodology's wording — peptides at $\geq 90\%$ are "prioritized as population-covering candidates" — below-target peptides were retained and ranked lower rather than deleted. Applying the criterion as a hard filter produced a construct containing **zero helper-T-cell epitopes**, since cumulative MHC-II coverage does not reach 90% at the individual-peptide level; such a construct would carry no GPGPG linkers, elicit no CD4+ help, and render the study's later immune-simulation research questions unanswerable.

1.9.2 Cumulative coverage

Per-epitope coverage answers what fraction of a population carries an allele binding one specific peptide. It is a prioritization signal, not the coverage of the vaccine. A recipient is covered if **any** epitope in the construct binds one of their alleles, so the operative figure is computed over the union of binding alleles across the whole epitope set — the same calculation IEDB's own Population Coverage tool performs for multi-epitope constructs, using the formula already specified in Section II.C.I.F.

The difference is large. Individually, only **4 of the 23 HLA-restricted epitopes** in the final construct clear 90%. Cumulatively:

Table 6.

Population coverage of the assembled construct

Measure	Coverage
---------	----------

Cumulative MHC-I (union of 65 alleles across 13 epitopes)	> 99.99%
Cumulative MHC-II (union of 32 alleles across 10 epitopes)	> 99.99%
Combined	> 99.99%

Table 7.

Per-locus coverage

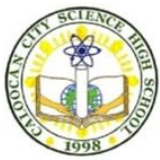
Locus	Coverage	Alleles bound
HLA-A	72.75%	16 / 22
HLA-B	97.04%	31 / 33
HLA-C	99.64%	18 / 19
HLA-DPB1	86.97%	8 / 8
HLA-DQB1	95.25%	6 / 7
HLA-DRB1	99.54%	18 / 18

The per-locus breakdown must be reported alongside the combined figure. The across-loci formula multiplies complements, so any three loci above roughly 85% drive the combined value past 99.9% almost regardless of the remainder; quoted alone, the combined number is close to uninformative. **HLA-A at 72.75% is the honest headline.** No locus saturates at 100%, confirming the result is not a computational artefact.

Population source and its limits. Allele frequencies were drawn from the Allele Frequency Net Database (AFND; Gonzalez-Galarza et al., 2020), population "**Singapore Riau Malay**" (pop_id 2042, n = 132, 107 two-field alleles across all six loci), retrieved 23 August 2026.

This is **Southeast Asian (Austronesian) proxy coverage, not Philippine coverage.** AFND holds no usable Philippine table for this analysis: the Philippines National Capital Region set (n = 51) — precisely the study's target population — is recorded at **one-field resolution only** (A*02 rather than A*02:01) and is therefore incompatible with two-field binding predictions; the only two-field Philippine dataset is "Philippines Ivatan" (n = 50), an indigenous Batanes population not representative of lowland Filipinos; and **HLA-DQB1 and HLA-DPB1 have no Philippine data at any resolution**, despite those two loci carrying most of the MHC-II coverage. Singapore Riau Malay was selected as the closest usable proxy on grounds of shared Austronesian ancestry and complete two-field data across all six loci.

The **strain** targeting of this study remains genuinely Philippine-specific: CRF01_AE and Clade II sequences were used throughout Phase I. It is the **HLA** dimension, and only that, which is a proxy.



1.10 Construct Assembly

The final construct, **Vax_Final_6f34b53e**, comprises 566 residues arranged N→C as follows:

Table 8.

Construct architecture

Region	Residues	Content
Adjuvant	1–45	Human β -defensin-3
Rigid linker	46–50	EAAAK
CTL cassette	51–204	13 MHC-I epitopes, AAY-joined
HTL cassette	205–404	10 MHC-II epitopes, GPGPG-joined
B-cell cassette	405–566	9 B-cell epitopes, KK-joined

The construct was selected from approximately **6,000 selection-and-ordering trials**, of which 60 candidates were polished and **10 reached zero bad hydrophobic windows and zero spurious linker motifs**.

Cross-class frame deduplication. The single most consequential selection rule in assembly is not stated in the current methodology and must be added. A candidate epitope sharing a sequence frame with anything already selected is skipped — **globally, across all target antigens and all epitope classes — and this check runs before any priority ranking**. The rule is correct: counting one antigenic region twice would inflate apparent antigen breadth and coverage. It is also the direct cause of the gp41 B-cell result reported below.

1.10.1 Antigen representation

Table 9.

Epitopes contributed per source antigen

Antigen	Epitopes
HIV-1 p24	7
Mpox A35R	6
Mpox B5R	6
Mpox L1R	6
HIV-1 gp41	5
HIV-1 gp120	2
HIV-1 p17	0 unique
Total	32

Seven antigens were screened but **six are uniquely represented**. HIV-1 p17 contributes no unique epitope: its two selected sequences are shared with p24, both being domains of the same Gag polyprotein. This is defensible on biological grounds and is stated here rather than claiming seven independently represented antigens.

Conservancy of the selected set: 17 of 32 epitopes are at 100% conservancy across their source variants, and none falls below 51.7%.

1.11 Experimental Corroboration in IEDB

Every epitope in this construct is computationally predicted. To distinguish predictions with prior laboratory support from genuinely novel candidates, all 32 construct epitopes were cross-referenced against IEDB's curated assay records via the IEDB Query API (19,193 curated records retrieved: HIV-1 977, Vaccinia 18,187, MPXV 29).

Table 10.

Experimental corroboration status

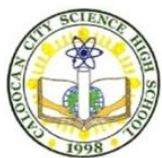
Category	Count
Experimentally confirmed (exact match or containment, with assay data attached)	1
Overlapping a curated epitope by ≥ 8 aa	16
Novel — no curated match	6
Corroborated total	26 / 32 (81%)

Assay classes represented across the corroborated set: MHC-ligand elution mass spectrometry, T-cell assays and B-cell assays. Species split: 12 same-species, 14 cross-species orthopoxvirus (vaccinia) orthologs. [VERIFY — the 12/14 split against the 10/16 confirmation tiers.]

Two qualifications must accompany this figure:

- Fourteen of the matches are cross-species.** IEDB holds 29 Monkeypox-virus epitopes against 17,903 vaccinia entries, and A35R, B5R and L1R have direct vaccinia orthologs. Orthopoxvirus cross-reactivity is precisely the mechanism by which smallpox vaccination confers protection against mpox, so restricting the analysis to MPXV alone would discard nearly the entire relevant experimental record. These matches are included but labelled CROSS_SPECIES and never silently merged with same-species evidence.
- Every match was verified against the study's own Phase 1A reference antigens** at 83–100% identity, with zero rejected. Sequence matching alone is insufficient — a 9-mer can align by chance within a roughly 200 kb poxvirus proteome — and protein names cannot settle it either, since genuine matches return under unified orthopoxvirus gene names such as "Ankyrin repeat protein OPG189" or "Protein OPG161," bearing no resemblance to "B5R" or "A35R." A name-based check would have rejected real orthologs.

The six novel epitopes are not a deficiency. Identifying epitopes with no prior curation is the purpose of a prediction



pipeline. This step distinguished the two categories; it did not rank them, and it added, removed or reordered no epitope in the construct.

Phase II: Structural Modelling and Verification

Table 11.

Phase II verdict summary

Step	Analysis	Verdict
2A	Physicochemical / toxicity / allergenicity	REVIEW
2B	Secondary and tertiary structure	COMPLETE
2C	Solubility and aggregation	REVIEW
2D	Stereochemistry (MolProbity)	VALIDATED
2D	Fold confidence → Phase III scope	DOMAIN_ONLY (residues 1–45)

No step rejected the construct, and no rejection reason was recorded at any step. Both review verdicts trace to a single underlying issue: predicted solubility.

2.1 Post-Construction Physicochemical Profile

Table 12.

Physicochemical properties of Vax_Final_6f34b53e

Property	Value	Criterion	Status
Length	566 aa	—	—
Molecular weight	~52 kDa	—	—
Instability index	39.23	< 40	PASS (stable)
GRAVY (Kyte–Doolittle)	−0.3219	≤ 0.4	PASS (hydrophilic)
Theoretical pI	9.70	outside pH 7.0 ± 0.5	PASS
Net charge	approx. +40	—	—
Hydrophobic windows > 80%	0	0 required	PASS
Spurious linker motifs	0	0 required	PASS
Junction cysteine violations (non-adjutant)	0	0 required	PASS

Junction-cysteine exemption — disclosed. The methodology mandates β -defensin-3 as the adjuvant and simultaneously rejects any 8–20 aa junction peptide carrying

more than two cysteines. Mature human β -defensin-3 natively contains **six cysteines** forming its three disulfide bonds; the two requirements cannot both be satisfied. The contradiction was resolved in the direction that preserves the adjuvant methodology requires: **three adjuvant-overlapping junctions are exempted and reported explicitly**, and it was verified that **zero non-adjuvant junctions violate the rule**. Retention of β -defensin-3 is supported by its established adjuvant activity (Tani et al., 2000).

2.2 Post-Construction Toxicity and Allergenicity

Test	Result
AllerTOP v2.0	NON-ALLERGEN
AllergenFP v1.0	NON-ALLERGEN
ToxinPred2 1.1	Pass
HemoPI2 1.3	0.72 — flag, not rejection (see below)
BLASTP vs UniProt Tox-Prot	Pass
Junction screen, 3,943 unique junction-spanning peptides	Pass
CamSol intrinsic solubility score	0.865041
DeepSol S2 solubility probability	0.4443739 — sole failing criterion

2.2.1 Hemolysis

HemoPI2 returned 0.72, above the 0.50 threshold. Under the methodology's conjunctive rule this is not a rejection, since that rule additionally requires a hydrophobic fraction above 0.60 and the measured value is 0.401 [VERIFY against the 0.409 reported at Step 2C]. The signal is nonetheless stated plainly: β -defensin-3 is a natural cationic antimicrobial peptide and is membrane-active by design, so a hemolysis signal from the adjuvant domain is biologically expected. This is the one flag in the study that would benefit most directly from experimental follow-up.

2.2.2 Junction composition flags

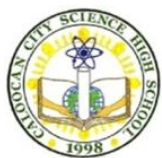
571 junction peptides carried composition-only toxicity flags with **zero** BLAST or MERCI evidence — exactly the false-positive mode the evidence-based toxicity rule was adopted to suppress. Recorded as advisory.

2.2.3 Secondary Structure

Table 13.

Secondary structure composition

Predictor	Helix	Sheet	Coil
-----------	-------	-------	------



PSIPRED	47.88%	5.12%	47.00%
NetSurfP-3.0	25.80%	4.06%	70.14%

NetSurfP-3.0 replaced RaptorX, whose server returns HTTP 500 and has not recovered. The two predictors disagree substantially on helix content and agree that β -sheet content is minimal, which is consistent with a construct composed of short unrelated epitopes joined by deliberately flexible linkers. DSSP assignment from the three-dimensional model (pydssp 0.9.1) is reported as a supplementary column.

2.4 Tertiary Structure and Fold Confidence

AlphaFold3 produced five ranked models. Because MolProbity scores local stereochemistry only — a random coil with clean geometry passes every one of its criteria — AlphaFold's own confidence output was analysed separately. This analysis is the most consequential single finding of Phase II.

Table 14.

Folded confidence by region

Metric	Adjuvant (1–45)	Epitope cassettes (51–566)
Mean pLDDT	91.3	21.4 – 39.7
Residues at pLDDT ≥ 70	100%	0%
Intra-region PAE	2.41 Å	22.1 – 29.0 Å
Cross-model C α RMSD (10 model pairs)	0.13 Å	18.5 – 23.9 Å

Whole-model metrics: **pTM 0.16–0.17 across all five ranked models; 84.6% of residues below pLDDT 50; only 46 residues reach pLDDT ≥ 70 , forming a single contiguous run (1–46)** corresponding exactly to the β -defensin-3 adjuvant; inter-region PAE approximately 31 Å against a ceiling of roughly 31.75 Å, meaning AlphaFold cannot place the cassettes relative to anything else in the model. [VERIFY — the confident run is reported as 1–46 while the architecture table defines the adjuvant as 1–45; state one boundary consistently.]

This is architecture, not modelling failure. Flexible GPGPG, AAY and KK linkers are chosen precisely so that each epitope remains independently accessible to antigen-processing machinery; a determined global fold is not the design goal and would arguably be a defect. No alternative epitope selection changes this — a materially different construct moved the equivalent measurements by approximately 0.01 — and molecular dynamics will not fold roughly 520 de novo residues.

2.4.1 Consequence for Phase III

Docking the whole model would mean docking one arbitrary conformer of a ~520-residue coil; substituting any other ranked model displaces those regions by 18–24 Å, so the result would not reproduce. Phase III is therefore scoped to residues 1–45. This is not a fallback: human β -defensin-3 is the TLR-1/2 and TLR-4 agonist and is the reason the adjuvant is in the construct at all (Tani et al., 2000). The epitope cassettes have no proposed receptor interaction to model. The region AlphaFold determines confidently is exactly the region on which the docking claim depends.

The thresholds applied here (pLDDT ≥ 70 , pTM ≥ 0.5 , cross-model RMSD ≤ 2.0 Å) are external conventions from the AlphaFold literature (Jumper et al., 2021; Krokidis et al., 2025); the methodology as written specifies no fold-confidence criterion.

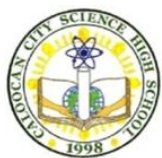
2.5 Solubility and Aggregation (Step 2C)

VERDICT: REVIEW

Table 15.

Solubility and aggregation results

Measure	Value	Threshold	Interpretation
DeepSol S2 probability	0.4443739	≥ 0.50 soluble	Predicted insoluble
CamSol intrinsic score	0.865041	$> +1$ highly soluble, < -1 poorly soluble	Soluble side
Aggrescan3D max patch	16 aa (res 230–245), mean pLDDT 22.0	> 8 aa flags	In a region AlphaFold cannot place
CamSol structural patch	7 aa (res 560–566), mean pLDDT 21.9	> 8 aa flags	Below threshold
Hydrophobic surface fraction	0.409	> 0.25 flags	See below
Exposed hydrophobic area	649.3 Å ²	> 250 Å ² flags	See below



FoldX total
folding energy 565.42
kcal·mol⁻¹ contextual Not a $\Delta\Delta G$

2.5.1 DeepSol

DeepSol S2 is a prediction, not a measurement, and it is contradicted by the other sequence-based predictor applied here: CamSol intrinsic scores the same sequence 0.865 — comfortably soluble, and the highest value of any construct generated in this project. Neither structure-aware tool corroborates insolubility: the Aggrescan3D patch (16 aa) and the CamSol structural patch (7 aa) both sit at mean pLDDT of approximately 22, in regions AlphaFold cannot place. Under the methodology's own compound rule in Section II.C.II.C — rejection requires DeepSol below 0.50 **and** a structure-aware tool identifying a patch above 8 contiguous residues in a pLDDT ≥ 70 region — the construct is **not rejected**, because no confident patch exists. Only wet-lab expression can settle the disagreement; no computational phase can. Length reduction does not resolve it: a 393-residue variant scored 0.393, worse than the longer constructs.

2.5.2 Hydrophobic surface flags

Both values exceed their thresholds, but both are computed by SASA over a model that is 84.6% disordered. They describe one arbitrary conformer of a flexible chain rather than a stable exposed surface. Calibration against solved structures using the identical FreeSASA code path supports this reading: RCSB TLR2 (6NIG) returns 0.203 and TLR4 (8WTA) returns 0.255, while this construct's own well-folded adjuvant domain returns 0.274 — essentially matching the solved-structure controls — and the disordered cassette region returns 0.428. The flag tracks model **confidence**, not epitope **composition**. It is reported as a disorder signature consistent with Section 3.2.4, not as an aggregation liability.

2.5.3 FoldX

The methodology lists "destabilizing $\Delta\Delta G$ predictions (FoldX web) $> +1.5$ kcal·mol⁻¹ for candidate surface mutations" as a rejection criterion. $\Delta\Delta G$ is by definition a difference between a mutant and a wild type, and **this study designs no mutants**, so the quantity has no referent. FoldX 5 was run locally with -command=Stability and the output is reported as total folding energy, contextual only.

2.6 Stereochemical Validation (Step 2D)

VERDICT: VALIDATED

Structure repair and minimisation (PDBFixer, OpenMM 8.6, Phenix 2.2) were performed **before** validation, so that MolProbity reports the quality of the coordinates actually carried forward rather than of a structure subsequently discarded.

Table 16.

MolProbity validation (criteria within goal)

Criterion	Value	Percentile / goal
MolProbity score	0.90	100th percentile; EXCELLENT band 0.5–1.5
Clashscore	0.68	99th percentile
Ramachandran favored	97.16%	target 95–98%
Ramachandran allowed	2.48%	—
Ramachandran outliers	0.35%	—
Rama distribution Z-score	-0.04 $\pm$ 0.34	—
Poor rotamers	1.08%	—
C β deviations	0.00%	—
Bad bonds	0 / 4,468	—
Bad angles	0 / 6,052	—
Chiral volume outliers	0 / 653	—

Table 17.

MolProbity validation (criteria outside goal)

Criterion	Value	Goal
Favored rotamers	91.54%	> 98%
Cis non-prolines	0.39% (2 of 518)	< 0.05%
C α geometry outliers	2.31% (13 residues)	< 0.5%
CaBLAM outliers	3.2% (18 residues)	< 1.0%

The four criteria in Table 3.14 do not enter the MolProbity score, which is computed from clashscore, rotamer outliers and Ramachandran outliers only, with built-in tolerances on the latter two. This is why a score of 0.90 does not reflect them, and it is why both tables must be reported.

The pattern is itself the finding. Every quantity Phenix restrains during refinement is perfect — bonds, angles, C β deviations, clashscore — while every quantity describing whether the backbone adopts a *believable conformation* is poor. Refinement produced ideal local geometry without producing a correct fold.

2.6.1 Localisation

Mapping molprobability.cablam output onto the per-residue pLDDT table, **59 of 62 flagged residues lie outside the adjuvant, at mean pLDDT 29.2** (whole model 35.7; adjuvant 91.3). All CaBLAM outliers fall in the epitope cassettes with none in the adjuvant, as do 12 of the 13 C α -geometry outliers, and they cluster in runs at residues 341–346 and 364–369 coinciding with GPGPG linker regions. [VERIFY — reconcile the 62-residue total against the individual counts in Table 3.14, and the "all 13 CaBLAM outliers" phrasing against the 18 reported.]

This constitutes independent corroboration of the fold-confidence finding from a tool built for exactly this situation: CaBLAM is MolProbity's low-resolution criteria section, designed for structures whose detailed geometry cannot be trusted. Four independent analytical routes — pLDDT, PAE, cross-model RMSD and CaBLAM — converge on the same conclusion.

For reference, the superseded 484-residue construct scored MolProbity 0.64 with clashscore 0.40 and 0.00% poor rotamers, so the final construct is marginally weaker on local geometry while remaining comfortably inside the EXCELLENT band.

3.0 Answers to Research Questions 1-3

RQ1 — Which epitopes satisfy antigenicity, conservancy, population coverage, non-toxicity and non-allergenicity?

Thirty-two epitopes: 13 MHC-I, 10 MHC-II and 9 B-cell, drawn from six uniquely represented source antigens (HIV-1 p24 7, MpoX A35R 6, MpoX B5R 6, MpoX L1R 6, HIV-1 gp41 5, HIV-1 gp120 2; HIV-1 p17 contributes no unique epitope). All 32 are non-toxic and non-allergenic and show no human self-homology. Seventeen are at 100% conservancy and none falls below 51.7%. Cumulative population coverage exceeds 99.99% for both MHC classes against a Southeast Asian Austronesian proxy population, with per-locus values ranging from 72.75% (HLA-A) to 99.64% (HLA-C). Twenty-six of the 32 carry curated experimental support in IEDB.

RQ2—What architecture ensures a stable construct with high physicochemical stability and favorable protein folding?

β-defensin-3 + EAAAK + 13 MHC-I epitopes (AAY) + 10 MHC-II epitopes (GP GPG) + 9 B-cell epitopes (KK), totalling 566 residues, selected from approximately 6,000 selection-and-ordering trials. Instability index 39.23 (stable, threshold 40), GRAVY -0.3219 (hydrophilic), theoretical pI 9.70 (well outside the buffer-pH exclusion zone), net charge approximately +40, with zero hydrophobic windows above 80%, zero spurious linker motifs and zero non-adjuvant junction-cysteine violations.

The phrase "favorable protein folding" requires **qualification**. The architecture is stable and well-formed at the sequence and local-geometry level. It does **not** adopt a determined global fold, and this is by design rather than by failure — see RQ3 and Section 3.2.4.

RQ3 — What is the stereochemical quality of the predicted tertiary structure, specifically regarding Ramachandran-favored regions?

97.16% Ramachandran-favored, inside the stated 95–98% target, with 0.35% outliers, a clashscore of 0.68 and a MolProbity score of 0.90, inside the stated 0.5–1.5 EXCELLENT band.

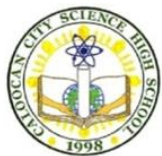
The same report returns 91.54% favored rotamers (goal > 98%), 2.31% α -geometry outliers (goal < 0.5%) and 3.2% CaBLAM outliers (goal < 1.0%), none of which enter the MolProbity score. **Stereochemistry is excellent; the fold is undetermined outside residues 1–46.** These are separate claims and Step 2D reports them as separate verdicts. RQ3 should be rescoped to ask about stereochemical quality **and fold confidence.**

CONCLUSIONS

[illegible]

INDIVIDUAL AUTHOR'S CONTRIBUTIONS

D.L.; Conceptualized the study design and methods, supervised manuscript revisions. J.P. and R.L.; Involved in exercise program formulation and revisions; suggested modifications in the study's methodologies. D.E.A.; Distributed the tasks, led submissions; cowrote the protocol, and performed manuscript revisions. D.E.A., J.B., E.J.C., S.A.F., J.J.L., D.K.R., K.S., and M.T.T.; Co-wrote the protocol and completed manuscript revisions. All read and approved the manuscript's final version. (Taken from: Lipardo, D., Añonuevo, D. E., Barlis, J., Chang, E. J., Ferrer, S. A., Lama, J. J., Rivera, D. K., Saludo, K., Turiano, M. T., Porto, J., & Laxa, R. (2022). Evaluation of an Arnis-based exercise program for community-dwelling older adults in the Philippines: An exploratory study protocol. *Philippine Journal*



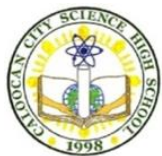
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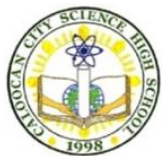
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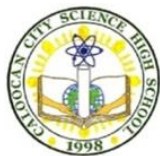
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